NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge Factsheet Adaptive Hierarchical-Degradation Super-Resolution via Omni-Paradigm Collaborative Learning

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1. Method Description

The motivation of our approach stems from two critical observations in real-world super-resolution (Real-SR). First, **degradation heterogeneity**: unlike synthetic datasets, real-world degradations are inherently stochastic and multi-faceted. They consist of a wide spectrum of complexities, ranging from simple isotropic blur to non-linear, high-order distortions and sensor-specific noise. The ill-posedness of the restoration task varies significantly across different degradation levels. Second, **distributional discrepancy**: there exists a profound data gap between the high-fidelity manifold sampled by generative models and the corrupted real-world manifold. Direct fine-tuning of a generative prior on complex SR tasks often forces the model to undergo a drastic weight shift, resulting catastrophic forgetting.

To mitigate these issues, we propose a Hierarchical Dual-LoRA structure. This staged adaptation framework ensures a smooth transition from the image generation to the SR task. We decompose the adaptation process into two sequential phases:

1. **Base-Adaptation Phase:** We first bridge the task gap via a *Base-Adaptation LoRA*, trained on mildly degraded synthetic data. This phase allows the model to learn the fundamental mechanics of reconstruction while retaining its high-quality generative priors.
2. **Refinement Phase:** We then address the "simple-to-complex" degradation gap using a *Refinement LoRA*. By freezing the base weights and optimizing the refinement module on complex artifacts, the model progressively learns to handle severe distortions without compromising structural integrity.

This hierarchical approach functions as a regularized bridge, effectively anchoring the model's generative prior while incrementally enhancing its robustness against real-world complexity.

Recognizing that different generative paradigms—such as Diffusion Models (DM), Flow Matching (FM), and Autoregressive (AR/VAR) models—exhibit complementary strengths in reconstructing diverse texture types, we propose an **Omni-Paradigm Teacher Distillation** strategy. During training, for a given low-quality input x_{LQ}, we generate a candidate set of pseudo-ground truths $\{\hat{x}_k\}_{k=1}^{K}$ from an ensemble of K pre-trained state-of-the-art teachers. We select the most suitable target by:

$$x_{target} = \arg\max_{\hat{x}_k} \mathcal{S}(\hat{x}_k, x_{HQ}) \tag{1}$$

where $\mathcal{S}(\cdot)$ is a multi-modal scoring function incorporating spatial fidelity, and perceptual quality. This ensures that the student model learns from the best teacher outputs.

Both LoRA modules are optimized using a comprehensive composite loss function to ensure balanced performance in fidelity and perception:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{MSE}} + \lambda_2 \mathcal{L}_{\text{LPIPS}} + \lambda_3 \mathcal{L}_{\text{GAN}} \tag{2}$$

2. Implementation Details

Our model is trained on the DIV2K and LSDIR datasets, as provided by the challenge organizers. We adopt the Real-ESRGAN degradation pipeline to synthesize LQ-HQ training pairs. We employ Stable Diffusion 2.1-base as our foundation model. For LoRA injection, we set the LoRA rank to 16 for the VAE and 32 for the UNet cross-attention and self-attention layers. The framework is optimized using the AdamW optimizer with a learning rate of 5×10^{-5} and a global batch size of 16. The model is first trained for $3,000$ steps on mild degradations to establish the fundamental super-resolution mapping. Subsequently, the model undergoes $5,000$ steps of refinement on complex, high-order degradations to enhance its restoration capabilities.

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Figure 1. Overview of the proposed framework. Our method features a Hierarchical Dual-LoRA architecture and an Omni-Paradigm Collaborative Learning strategy.